RAPID PEDS DOSING GUIDE

RESPIRATORY

Inhalers/Nebulization ~ Add 2.5-3 ml of NS to nebulized meds

Salbutamol [Ventolin]

Inhalation		Nebulization	
5 - 10 kg	4 puffs/dose	0.15 mg/kg/dose (max 5mg/ dose)	
10 - 20 kg	6 puffs/dose	<15kg / <5yrs	2.5 mg/dose = 0.5ml/dose
> 20 kg	8 puffs/dose	>15kg / >5yrs	5 mg/dose = 1ml/dose

Ipratropium bromide [Atrovent]

Inhalation		Nebulization	
< 5 yrs	2 puffs/dose	< 5 yrs	250 mcg/dose
5 - 12 yrs	4-8 puffs/dose	5 - 12 yrs	250 - 500 mcg/dose
> 12 yrs	8 puffs/dose	> 12 yrs	500 mcg/dose

Budesonide [Pulmicort]

< 4 years		5 - 11 years	
Low dose	0.25-0.5 mg/day	Low dose	0.5 mg/day
Med. dose	0.5 - 1 mg/day	Med. dose	1 mg/day
High dose	1 mg/day	High dose	2 mg/day

Epinephrine (L-epinephrine)

Neb 0.5 ml/kg/dose of 1:1000 solution (max 5 ml/dose) diluted in NS

< 20 kg	2.5 ml in 2.5 ml NS	> 20 kg	5 ml in 2.5 ml NS

Flixotide	(> 4 years) 125 mcg 1 puff BD
3% NaCl	2-3 ml/dose (up to Q6hr)

Steroids

Prednisolone (medium potency) [Asthma]

< 12 years	PO 1-2 mg/kg/day OD or BD (max 60mg/day)
> 12 years	PO 40-60 mg OD or BD

RESPIRATORY

Steroids

Methylprednisolone (medium potency) [Asthma]	
PO / IM / IV	1-2 mg/kg/day Q 6-8h (max 60mg/day)

Hydrocortisone	
PO	2.5-10 mg/kg/day or 75-300 mg/m2/day Q6-8h
IM/IV	0.56-8 mg/kg/day or 20-240 mg/m2/day Q6-8h
Replacement	(limited use) 1-5 mg/kg/day or 30-150 mg/m2/day BD / OD

Dexamethasone (High potency)		
Anti-inflammatory	0.02 - 0.3 mg/ kg/ day Q 6-12h	
Asthma	0.3-1.7 mg/kg/dose OD for 1-2 days (max 16 mg/dose)	
Airway Edema	0.5mg/kg/dose Q6-12h before extubation (max 10mg/dose) then Q6h for 6 doses (total 3mg/kg)	
Croup	Mild = low dose	0.15-0.6 mg/kg OD (max 16-20 mg/dose)
	Mod-severe = high dose	0.6 mg/kg (max 16-20 mg/dose)
MgSO4	**25-75 mg/kg/dose infused over 15-60 min +30 ml NS (max 2g/dose)**	

CARDIOVASCULAR

Adenosine (Rapid IV)		
HD stable	**< 50 kg**	Rapid IV 0.05-0.1 mg/ kg (max 6mg/dose)
	> 50 kg	6 mg / dose
HD unstable	**1st dose**	0.1 mg/kg (max 6mg)
	2nd dose	0.2mg/kg (max 12 mg/dose)
Amiodarone	IV 5 mg/kg (max 300mg)	
Epinephrine	IV 0.01 mg/kg (0.1 ml / kg of 0.1 mg/ ml solution)	
PGE1	IV 0.05-0.1 mcg/kg/min	
Atropine	IV 0.02 mg/kg (max 0.5 mg/dose)	

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CNS

(Infusion doses are not added - to be given under neurology / PICU care)
***maintenance dose usually starts 12 hours after loading dose**

Lorazepam [Ativan]	
IV	0.1 mg/kg (max 4 mg/dose)
Midazolam [Dormicum]	
IV / IO	0.1 - 0.3 mg/kg (max 10 mg)
IM	0.1 - 0.2 mg/kg (max 10 mg)
Buccal	0.2 - 0.5 mg/kg (max 10 mg/dose)
Diazepam [Valium]	
IV / IO	0.2 - 0.3 mg/kg (max < 5yrs: 5 mg/dose, max > 5yrs: 10 mg)
Rectal	0.5 mg/kg (max 20 mg)
Levetiracetam [Keppra]	
Loading	30 - 60mg/kg (max 4.5g/dose) over 5-15 mins diluted in NS
Maintenance	10 mg/kg/dose BD - increase 5mg every 3 days 'til 30 mg/kg/day
Phenobarbitone	
Loading	15 - 20mg/kg diluted in 50 ml NS infused slowly over 20 mins (max 1g/dose)
Maintenance	3-6 mg/kg/day twice daily
Phenytoin	
Loading	20 mg/kg diluted in 50 ml NS infused slowly over 20 min (max 1.5 g/dose)
Maintenance	5 mg/kg/day twice daily (max 300 mg/day)
Na Valproate [Depakine]	
Loading	20 - 40 mg/kg over 5 mins diluted in NS or D5W (max. 3g)
Maintenance	Initially 10mg/kg/day BD - increase 5mg/kg/day per week (till target) OR divide therapeutic dose to be increased Q3days (over 10-14 days till target)

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MISCELLANEOUS

Paracetamol	10 - 15 mg/kg/dose (max 75 mg/kg/day OR 4 g/day)			
	4 hourly regimen	10 mg/kg/dose	**6 hourly regimen**	15 mg/kg/dose
Brufen (>6 months, >8 Kg)	4-10 mg/kg/dose (max 40 mg/kg/day or 2.4 g/day whichever less) ***Do not use more than 4 doses a day***			
Zyrtec	**< 5 years**	2.5 mg OD/BD	**> 5 years**	5 - 10 mg OD
Losec	1 mg/kg/dose (max 40mg/day)			
Zofran	0.15 mg/kg/dose (max 16 mg)			
Lasix	1 - 2 mg/kg/dose			
Morphine (Opioid)	**< 50kg or < 12yrs**	IV 0.05 - 0.1mg/kg/dose (max 4 mg/dose, not more than every 4 hours		
	> 50kg or > 12yrs	IV 2-5mg (not more than every 4 hours)		
Ketamin (Dissociative)	**Don't use < 3months** IV 1 - 2 mg/kg (can repeat every 5 - 15 min, max 4 mg/dose)			
Vitamin D	**Prophylactic**	< 1 year: 400 IU OD | >1 year: 600 IU OD		
	Therapeutic	2000 IU for 6 weeks (Rickets: precede by 1 week Ca+)		
Iron	**Prophylactic**	1 mg/kg/day		
	Therapeutic	3-6 mg/kg/day divided by BD/TDS		
pPRBC	10 - 15 ml/kg over 4 hrs + with Lasix (mid or post or mid+post transfusion)			
PLT	10 ml/kg slowly = 1 unit/5 kg (over 30-60 mins | large volume: infused over 2 hrs)			
FFP	10 - 15 ml/kg slowly (over 30-60 mins, not more than 4 hrs)			

Maximum IVF rate is 100ml/hr = 2400 ml/day

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ANTIBIOTICS

Beta-Lactams

Amoxicillin

Standered PO dose	40-45 mg/kg/day BD / TDS (max: 500mg/dose, 1.5g/day)
High PO dose	80-90 mg/kg/day BD or TDS (max: 4 g/day)
Neonatal dose	50 mg/kg/dose BD

Augmentin (Amox-Clav)

Standered PO dose	40 - 45 mg/kg/day (PO: divided BD | IV: divided TDS)
High PO dose	80 - 90 mg/kg/day (PO: divided BD | IV: divided TDS) (max 1.2g /dose)
Neonatal IV dose	50 mg/kg/dose (< 1 week age: BD | > 1 week age: TDS)

Ampicillin (IV)

Mild infections	50-100 mg/kg/day Q6h
Anti-septic dose	150 mg/kg/day Q6h
Anti-meningetic / Pneumonia	200 mg/kg/day Q6h (max: 8 g/day)
CNS infection	300 - 400 mg/kg/day Q6h (max: 12 g/day)

3rd Genertion Cephalosporins

Cefixime [Suprax]

PO	8 mg/kg/day OD or BD (max 400 mg/day)

Ceftriaxone [Rocephin]

Standard dose	IV/IM 50-75 mg/kg/day OD or BD (max 2g/day)
Anti-meningetic	IV 100 mg/kg/day OD (max 4g/day)

Cefotaxime [Claforan] IV

Standard dose	100 mg/kg/day Q6h	Anti-septic	150 mg/kg/day Q6h
Anti-meningetic	200 mg/kg/day Q6h	CNS infection	300 mg/kg/day Q6h (Max 12g/day)
Neonatal dose	50 mg/kg/dose <1 week age: BD | > 1week age: TDS | meningitis: Q6h		

ANTIBIOTICS

Macrolides

Clarithromycin [klacid]

PO	7.5mg/kg/dose BD (max 500mg/dose, 1g/day)

Lincosamide

Clindamycin

Skin/soft tissue infection	PO/IV 10-25 mg/kg/day divided TDS (max: 1.8 g/day)
Anaerobic infection	PO/IV 20-40 mg/kg/day divided Q6h or TDS (max 2.7 g/day)
Otitis media Toxic infection /NEC	IV 40 mg/kg/day divided TDS or Q6h
Neonatal dose	5 mg/kg/dose TDS

Tazocin [Piperacillin/Tazobactam]

Standered dose	IV 240 - 300 mg/kg/day Q6-8h = [100mg/kg/dose TDS] (max: 16 g/day)
Severe infection	80 mg/kg/dose Q6h
Neonatal dose	100 mg/kg/dose (< 1week age: BD | >1 week age: TDS)

Aminoglycosides

Gentamicin

[Check trough 30 mins before next dose (N <1 mcg/ml)]

IV/IM	2 - 2.5 mg/kg/dose TDS or 5 - 7.5 mg/kg/day OD			
Neonate IV/IM	< 1 week age	5mg/kg OD	> 1 week age	7.5 mg/kg OD

Amikacin

[Check trough 30 mins before next dose (N <5 mcg/ml)]

Standered dose	IV 7.5 mg/kg/dose BD/TDS or 15-20 mg/kg OD (max 20 mg/kg/day)			
Anti-meningetic	IV 20 mg/kg/day divided TDS			
Neonate IV/IM	< 1 week age	15 mg/kg OD	> 1 week age	20 mg/kg OD

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ANTIBIOTICS

Glycopeptides

Vancomycin
[Check trough 30 mins before 4th dose]
(N: 10-15 mcg/ml in mild infection, N: 15-20 mcg/ml in severe infection)

PO	5-10 mg/kg/dose Q6h (max 2g/day)
IV	15 mg/kg/dose q6-8h (max 2-4 g/day)
Neonatal IV dose	15 mg/kg/dose (< 1 week age: BD | > 1 week age: TDS)

Carbapenem

Meropenem

Standard IV dose	20 mg/Kg/dose TDS (max 1g/dose)
Severe Infection Anti-meningetic dose	40 mg/Kg/dose TDS (max 2g/dose, 6g/day)
Neonatal IV dose	20 - 40 mg/kg/dose (< 1 week age: BD | > 1 week age: TDS)

ANTIVIRALS

Oseltamivir [Tamiflu]
(clinical benefit is greatest if initiated within 48 hours) for 5 days

	Infants		Children	
Therapeutic	**< 38 wks**	1 mg/kg/dose BD	**< 15 Kg**	30 mg BD
	38 - 40 wks	1.5 mg/kg/dose BD	**15 - 23 Kg**	45 mg BD
	40 wks - 9 months	3 mg/kg/dose BD	**23 - 40 kg**	60 mg BD
	> 9 months	3.5 mg/kg/ dose BD	**> 40 Kg**	75 mg BD
Prophylactic	Limited Data available		Similar to the above (but OD)	

Acyclovir [For HSV Encephalitis]

IV	10-15 mg/kg/dose TDS x 21 days (children 4 months - 12 yrs)

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ANAPHYLAXIS

Epinephrine	
IM	0.01 ml/kg/dose or 0.01mg/kg/dose (1mg/ml solution) (max 0.3 mg/dose)
IV	(Refractory) IV infusion (1 mcg/ml) : 0.1-1 mcg/kg/min (max 10 mcg/min)
Diphenhydramine (Not first line - adjunct to Epinephrine)	
IV/IM/PO 1 mg/kg/dose (max 50 mg/dose) over 5 mins , may repeat Q6-8hrs	

PEDIATRIC EQUATIONS

NGT size	[Age + 16] / 2			
ETT size	Cuffed	[Age /4] + 3.5	UnCuffed	[Age /4] +4
Minimum sBP	70 + [2 x Age in yrs]			
Avg. Weight	8 + [2 x Age in yrs]			
BSA	$\sqrt{\frac{height \times weight}{3600}}$			
QTc	$\frac{QT}{\sqrt{RR}}$	one small square = 0.04s		
Insulin Sensitivity Factor (ISF)	100/ TDS			
Insulin Carb. Ratio (ICR)	300-500 / TDD			
Insulin Correction Dose	$\frac{\text{Current RBS} - \text{Target RBS}}{\text{ISF}}$			
GIR	$\frac{\text{D\% infused} \times \text{infusion rate (ml/kg/day)}}{\text{weight (kg)} \times 6}$			